Supplementary Material for:Temporal decoupling of CTLA-4 and ERK sensitivity in brain tumour signaling: a theoretical basis for sequencing checkpoint and MAPK inhibition

Table S1: Kinetic Parameters

Table 1: Numerical values for kinetic parameters used in modeling.

S.No	Kinetic Parameters	Value	Unit	References
1	$k^{+}_{\text{CTLA-4_CD80}}$	1×10^{5}	$\text{M}^{-1}\,\text{s}^{-1}$	Estimated
2	$k^{-}_{\text{CTLA-4_CD80}}$	0.43	s^{-1}	[1]
3	$K^{\text{PP2A}}_{\text{cat CTLA-4_CD80}}$	0.01	$\text{M}\,\text{s}^{-1}$	Estimated
4	$K^{\text{PP2A}}_{\text{m CTLA-4_CD80}}$	1×10^{-5}	M	Estimated
5	$k^{+}_{\text{PP2A_ERK}}$	0.01	$\text{M}^{-1}\,\text{s}^{-1}$	Estimated
6	$k^{+}_{\text{RTKs_Grb2}}$	53	$\text{M}^{-1}\,\text{s}^{-1}$	[2]
7	$k^{-}_{\text{RTKs_Grb2}}$	4.3×10^{-5}	s^{-1}	[2]
8	$k^{+}_{\text{PI3K_RTKs}}$	0.001	$\text{M}^{-1}\,\text{s}^{-1}$	[3]
9	$k^{+}_{\text{RTKs_NF2}}$	0.01	$\text{M}^{-1}\,\text{s}^{-1}$	Estimated
10	$k^{+}_{\text{RTKs_Grb2_SOS}}$	1.6×10^{5}	$\text{M}^{-1}\,\text{s}^{-1}$	[4]
11	$k^{-}_{\text{RTKs_Grb2_SOS}}$	0.02	s^{-1}	[4]
12	$k^{+}_{\text{Ras-GTP_RTKs_Grb2_SOS}}$	0.001	s^{-1}	Estimated
13	$k^{+}_{\text{Raf_Ras-GTP}}$	3.6×10^{7}	$\text{M}^{-1}\,\text{s}^{-1}$	[5]
14	$k^{+}_{\text{Ras-GTP_PP2A}}$	0.01	$\text{M}^{-1}\,\text{s}^{-1}$	Estimated
15	$k^{+}_{\text{Ras-GTP_NF2}}$	0.01	$\text{M}^{-1}\,\text{s}^{-1}$	Estimated
16	$k^{+}_{\text{PI3K_Ras-GTP}}$	0.001	s^{-1}	Estimated
17	$k^{+}_{\text{MEK_PP2A}}$	0.01	$\text{M}^{-1}\,\text{s}^{-1}$	Estimated
18	$k^{-}_{\text{MEK_PP2A}}$	0.001	s^{-1}	Estimated
19	$k^{+}_{\text{ERK_PP2A}}$	0.01	$\text{M}^{-1}\,\text{s}^{-1}$	Estimated
20	$k^{+}_{\text{MEK_ERK}}$	5.49×10^{8}	$\text{M}^{-1}\,\text{s}^{-1}$	[6]
21	$k^{-}_{\text{CyclinD_ERK}}$	0.001	s^{-1}	Estimated
22	$k^{+}_{\text{p16_ERK}}$	0.001	$\text{M}^{-1}\,\text{s}^{-1}$	Estimated
23	$k^{-}_{\text{ERK_PP2A}}$	0.01	s^{-1}	Estimated
24	$k^{+}_{\text{p27_AKT}}$	0.01	$\text{M}^{-1}\,\text{s}^{-1}$	Estimated
25	$k^{-}_{\text{CDK1_PP2A}}$	0.001	s^{-1}	Estimated
26	$k^{+}_{\text{RB_SMO}}$	1×10^{3}	s^{-1}	Estimated
27	$k^{+}_{\text{AKT_PTEN}}$	0.01	$\text{M}^{-1}\,\text{s}^{-1}$	Estimated
28	$k^{+}_{\text{AKT_PP2A}}$	0.01	$\text{M}^{-1}\,\text{s}^{-1}$	Estimated
29	$k^{+}_{\text{AKT_PDK1}}$	0.001	$\text{M}^{-1}\,\text{s}^{-1}$	Estimated
30	$k^{-}_{\text{TOR_AKT}}$	0.01	s^{-1}	Estimated
31	$k^{+}_{\text{TOR_NF2}}$	0.001	$\text{M}^{-1}\,\text{s}^{-1}$	Estimated
32	$k_{\text{CyclinD_CDK4}}$	1×10^{5}	$\text{M}^{-1}\,\text{s}^{-1}$	[7]

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S.No	Kinetic Parameters	Value	Unit	References
33	$K_{\text{CyclinD_CDK4}}$	7×10^{-6}	M	[7]
34	$k^{-}_{\text{RB_p16}}$	0.01	s^{-1}	Estimated
35	$k_{\text{RB_CyclinD_CDK4}}$	0.001	s^{-1}	Estimated
36	$k^{+}_{\text{RB_CyclinE_CDK2}}$	0.01	$\text{M}^{-1}\,\text{s}^{-1}$	Estimated
37	$k^{-}_{\text{RB_E2F}}$	0.001	s^{-1}	Estimated
38	$k^{+}_{\text{RB_E2F}}$	0.001	$\text{M}^{-1}\,\text{s}^{-1}$	Estimated
39	$k^{+}_{\text{p14_E2F}}$	0.001	$\text{M}^{-1}\,\text{s}^{-1}$	Estimated
40	$K_{\text{p14_E2F}}$	1×10^{5}	M	[8]
41	$k^{+}_{\text{p14_E2F}}$	1×10^{5}	M	[8]
42	$k^{+}_{\text{p53_p21}}$	0.01	$\text{M}^{-1}\,\text{s}^{-1}$	Estimated
43	$k^{+}_{\text{p53_MDM2}}$	0.01	$\text{M}^{-1}\,\text{s}^{-1}$	Estimated
44	$k^{+}_{\text{CyclinE_CDK2}}$	1×10^{5}	$\text{M}^{-1}\,\text{s}^{-1}$	Estimated
45	$k^{-}_{\text{CyclinE_CDK2}}$	0.01	s^{-1}	Estimated
46	$k^{+}_{\text{CDK2_p21}}$	0.01	$\text{M}^{-1}\,\text{s}^{-1}$	Estimated
47	$k^{+}_{\text{CyclinE_p27}}$	0.01	$\text{M}^{-1}\,\text{s}^{-1}$	Estimated
48	$k^{+}_{\text{p14_p53}}$	0.01	$\text{M}^{-1}\,\text{s}^{-1}$	Estimated
49	$k^{+}_{\text{MDM2_p14}}$	0.01	$\text{M}^{-1}\,\text{s}^{-1}$	Estimated
50	$K^{\text{MDM2}}_{\text{catp53}}$	0.001	$\text{M}\,\text{s}^{-1}$	Estimated
51	$K^{\text{MDM2}}_{\text{mp53}}$	1×10^{-6}	M	Estimated
52	$k^{+}_{\text{p21_p53}}$	0.001	s^{-1}	Estimated

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